Golden Sniffer: Low-Complexity 5G New Radio Signal Interception Exploiting Gold Sequences

Ivan Palamà
CNIT
Roma, Italy

Stefano Mangione*
Università degli Studi di Palermo
Palermo, Italy

Alessandra Dino*
Università degli Studi di Palermo
Palermo, Italy

Giulia Focarelli*
University of Rome Tor Vergata
Rome, Italy

Giuseppe Bianchi*
University of Rome Tor Vergata
Rome, Italy

Abstract

Sniffing 5G New Radio (NR) transmissions presents a significantly greater challenge compared to LTE, primarily due to the concealment of Radio Network Temporary Identifiers and relevant scrambling parameters. Existing techniques rely on exhaustive searches and simplifying assumptions that often do not hold in real deployments. In this paper, we propose a fundamentally different approach that exploits the linearity and non-cryptographic nature of the Gold sequences used in 5G NR for scrambling, drastically reducing the sniffer parameter space to a handful of test alternatives. We also propose and assess a countermeasure that disrupts the phase-based demodulation reference signal reconstruction used by our approach, with only minor impact on legitimate users.

CCS Concepts

• **Security and privacy → Mobile and wireless security**; **Privacy-preserving protocols**.

Keywords

5G, Sniffing, Security, Confidentiality, Air Interface

ACM Reference Format:
Ivan Palamà, Stefano Mangione, Alessandra Dino, Giulia Focarelli, and Giuseppe Bianchi. 2026. Golden Sniffer: Low-Complexity 5G New Radio Signal Interception Exploiting Gold Sequences. In *The 32nd Annual International Conference on Mobile Computing and Networking (MobiCom '26), October 26–30, 2026, Austin, TX, USA.* ACM, New York, NY, USA, 16 pages. https://doi.org/10.1145/3795866.3796673

*Also with CNIT, laboratories in Palermo and Rome, Italy

This work was supported by the European Union under the Italian National Recovery and Resilience Plan (NRRP) of NextGenerationEU (PE00000014 SERICS and PE00000001 RESTART).



This work is licensed under a Creative Commons Attribution 4.0 International License.

MobiCom '26, Austin, TX, USA
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2505-0/2026/10
https://doi.org/10.1145/3795866.3796673

1 Introduction

5G New Radio introduces a highly dynamic physical layer architecture, including user-specific Bandwidth Parts (BWPs), variable control channel formats, and complex Demodulation Reference Signal (DM-RS) schemes to support diverse applications [1–3]. This increased PHY-layer complexity is often interpreted as providing implicit protection, fostering the perception [4–6] that passive sniffing of control and data traffic is significantly more challenging in 5G NR than in Long Term Evolution (LTE).

Why 5G sniffing is (much) harder than 4G. In LTE, Physical Downlink Control Channel (PDCCH) and Physical Broadcast Channel (PBCH) control information follows fixed mappings and uses parameters sent unencrypted [7], enabling straightforward decoding. In 5G NR, critical configuration is conveyed through either scrambled (pre-security) or encrypted (post-security) Radio Resource Control (RRC) messages, making passive interception significantly harder. More specifically, a passive observer lacks knowledge of two user-specific temporary parameters essential for sniffing: (i) a dynamically assigned 16-bit Radio Network Temporary Identifier (RNTI) that uniquely identifies each receiver during an access session, and (ii) an additional 16-bit identifier used for "scrambling".

Regarding (i), a passive observer may extract the RNTI during the Random Access procedure, as done by State-of-the-art (SOTA) 5G sniffers such as Sni5Gect [8] and NR-Scope [9]. However, unlike in 4G LTE, this is the only opportunity, as the RNTI is no longer transmitted in clear after Random Access. The only publicly available sniffer that does not rely on unencrypted UE registration is 5GSniffer [5], which instead resorts to computationally intensive methods. In contrast, our approach eliminates brute-force searches by blindly deriving this (and the subsequent) key parameter on the fly, substantially improving efficiency.

A similar challenge concerns parameter (ii), the *Scrambling ID*, a 16-bit value initializing the pseudo-random sequence used before modulation for interference reduction and user separation. While it defaults to the Physical Cell ID (PCI), our experiments show that most real-world deployments typically reassign it dynamically after authentication. As a result, it must also be blindly decoded, and the method in [5], though effective, remains computationally expensive.

Finally, each RNTI/Scrambling ID pair requires testing multiple combinations of additional parameters, namely five Aggregation Levels (ALs), three possible Downlink Control Information (DCI) durations, and up to 100 candidate sizes, further increasing the complexity of passive interception.

Harnessing Gold Sequences. Instead of relying on carefully engineered brute-force optimizations [5], our approach follows a *novel and fundamentally different strategy*: it exploits the *non-cryptographic* structure of the Gold (pseudo-random) sequences used for scrambling, hence the name *Golden Sniffer*. Specifically, we leverage the fact that CRC attachment, RNTI masking, and scrambling are *linear over GF(2)*, which induces a modulo-2 linear dependence of the scrambled bits on the DCI payload, the RNTI, and the Scrambling ID. By capturing transmitted DM-RS sequences, we *directly recover their initialization parameters* with $O(1)$ *computational complexity* after an initial caching and pruning phase. Moreover, while [5] is limited to PDCCH decoding, our sniffer extends to single-layer Physical Downlink Shared Channel (PDSCH), as it does *not* rely on higher-layer configurations. This enables *full downlink (DL) decoding*, making our approach more versatile.

Countering our sniffer. Our approach recovers the PDCCH-embedded DM-RS by analyzing frequency-domain phase estimates and exploiting the limited constellation of the Quadrature Phase Shift Keying (QPSK) modulation. A natural countermeasure is therefore to disrupt passive phase estimation while preserving reliable reception for legitimate users, which possess knowledge of the RNTI and Scrambling ID. As a proof of concept, we propose a simple yet effective strategy: modulating the PDCCH frequency-domain amplitude with a cosine-squared profile to introduce controlled deep fading. Preliminary experimental results show that legitimate users incur only minor degradation (a few decibels), while passive phase estimation becomes unreliable, preventing recovery of the sequence parameters. We believe these findings might point to a potential design direction for 6G PHY layers with built-in resilience to eavesdropping.

Contribution. In summary, our contribution is threefold:

- We introduce a novel method leveraging Gold sequence properties to simplify blind decoding of critical parameters for passive 5G NR sniffing, without relying on clear-text registration messages;
- We design and implement *Golden Sniffer*, a tool enabling both control (PDCCH) and data (PDSCH) decoding, and validate its operation in both lab and real-world networks;
- We propose and experimentally assess a PHY-layer countermeasure, which induces targeted deep fading to disrupt unauthorized phase reconstruction while minimally impacting user performance.

The remainder of this paper is organized as follows. Section 2 reviews prior work on LTE and 5G NR sniffers. Section 3 outlines the challenges inherent in passive monitoring. Section 4 presents our *Golden Sniffer*'s approach and design, while Section 5 assesses its operation in controlled and real-world environments. Section 6 analyzes the proposed countermeasure to our approach, whereas Section 7 discusses its limitations and future research directions. Finally, Section 8 concludes the paper.

2 Related Work

Cellular networks, since the earliest generations and continuing with 5G, have historically relied on explicit security mechanisms, such as over-the-air encryption and temporary identifiers (e.g., Temporary Mobile Subscriber Identity (TMSI), Globally Unique Temporary Identifier (GUTI), and RNTIs), to protect user privacy. Despite these measures, vulnerabilities persist: 4G LTE is susceptible to PHY-layer attacks including jamming and spoofing [17–20], fake base stations [21–25], man-in-the-middle [26], and overshadowing [27, 28]. Moreover, despite 3GPP's security enhancements, these threats are only partially mitigated, leaving 5G NR vulnerable to analogous attack vectors [6, 29–33].

Regarding sniffing, commercial tools such as Keysight's WaveJudge [34], Qualcomm's QXDM [35], Actix Analyzer [36], TEMS™ Investigation [37], and thinkRF's Autonomous Spectrum Intelligence Platform [38] support LTE and emerging networks but are costly and closed-source, limiting research accessibility. For this reason, academic efforts have developed various air-interface monitoring tools over the past decade; Table 1 summarizes key LTE and NR solutions.

LTE sniffers. LTE sniffing has been addressed by several works. Kumar et al. [10] introduced LTEye, one of the first systems to demonstrate mobile network sniffing using commercial-off-the-shelf (COTS) Software-Defined Radio (SDR) without operator support. However, its candidate DCIs re-encoding verification is highly sensitive to interference, noise, and synchronization errors, thus requiring near-ideal radio conditions. RMon [11] combines control channel decoding with PHY-layer monitoring to infer network load and cell utilization, but its reliance on ideal conditions limits practicality. To overcome re-encoding limitations, C3ACE [12] uses short-term histograms of decoded RNTIs to filter out false positives, improving robustness at the cost of delayed detection for

Table 1: Comparative Summary of Related Work on 4G/5G Sniffers.

Work	Mobile Network Technology	BWP Support[1]	Physical Layer channels[2]		Complete DCI Decoding	Blind Decoding
			PDCCH	PDSCH		
Kumar et al. [10]	4G LTE	N/A	●	○	✓[a]	✓
Xie et al. [11]	4G LTE	N/A	●	○	✓	✓
Falkenberg et al. [12]	4G LTE	N/A	●	○	✓[b]	✓
Bui et al. [13]	4G LTE	N/A	●	◐ (RAR)	✓[b]	✓
Falkenberg et al. [14]	4G LTE	N/A	●	○	✓	✓
Kotuliak et al. [15]	4G LTE	N/A	●	●	✓	✓
Hoang et al. [16]	4G LTE	N/A	●	●	✓	✓
Ludant et al. [5]	5G NR	✗	●	○	✗[c]	✓
Richards et al. [4]	5G NR	✗	●	○	✗[c]	✓
Wan et al. [9]	5G NR	✓	●	○	✓	✗
Leo et al. [8]	5G NR	✓	●	●	✓	✗
Our Work	**5G NR**	✓	●	●	✓	✓

[1] Indicates support for network configurations with UE-dedicated BWP, allowing adaptation to multiple BW configurations within a single carrier.
[2] Indicates the physical layer channels investigated in the study: ●: Decoded, ◐ : Partially decoded, ○: Not decoded.
[a,b] Requires good signal quality and/or long monitoring time.
[c] Assume PCI as Scrambling ID (default configuration).

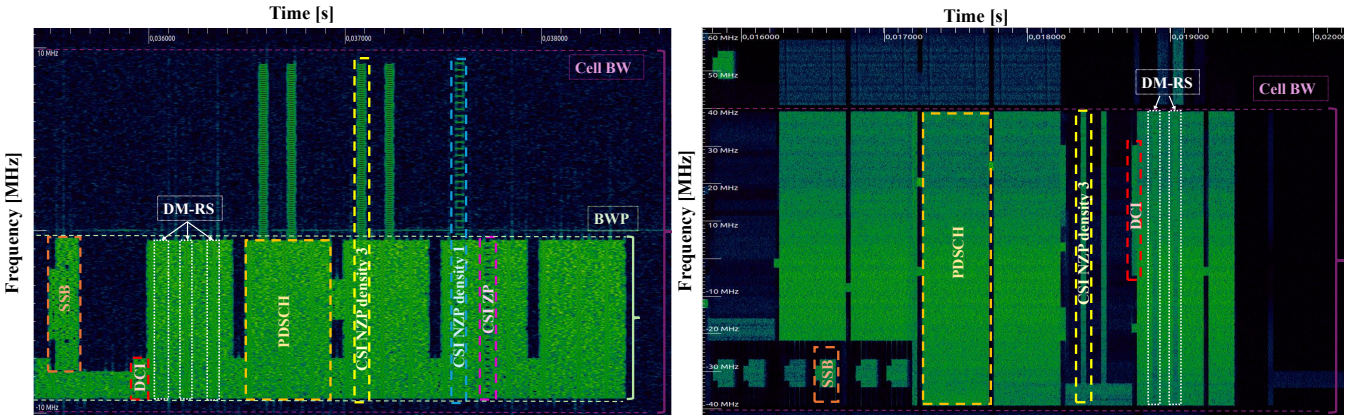
infrequently scheduled User Equipments (UEs). OWL [13] tracks initial RNTIs from Random Access Response (RAR) messages and applies coherence checks to exclude invalid DCIs; while it fully decodes the PDCCH and partially the PDSCH, its reliance on re-encoding reduces reliability under poor conditions. FALCON [14] reduces false detections using a recursive, depth-first shortcut-decoding method. LTEProbe [15] enables passive user tracking by correlating uplink (UL) and DL transmissions and leveraging Timing Advance for precise location estimation. LTESniffer [16] is an open-source tool that decodes both DL and UL traffic, infers user-specific configurations and compensates for propagation delays.

5G NR sniffers. With 5G NR, sniffers must rely on complex blind decoding techniques, often resulting in low DCI detection rates, or must capture the very first clear-text registration exchanges to extract configuration hints. 5GSniffer [5], the first open-source tool targeting 5G NR control channels, demonstrates the feasibility of blind extraction of PDCCH information. However, it fails to achieve fully reliable DCI decoding [39, 40], is limited to single-symbol DCIs, and lacks support for flexible BWP allocation. Similarly, Richards et al. [4] have shown promising results in extracting user RNTIs from the PDCCH. Both studies make the simplifying assumption that the Scrambling ID equals the PCI, thereby setting the RNTI used in the scrambling process (i.e. RNTI_{SCR} in Sec. 4.3) to zero instead of the actual RNTI used to mask the Cyclic Redundancy Check (CRC), as found in commercial deployments (Sec. 5.3). This assumption reduces the brute-force sniffer PDCCH DM-RS cardinality from 2^{32} to 2^{16}, making the proposed approaches feasible.

More recently, NRScope [9] demonstrated practical PDCCH decoding, and Sni5Gect [8] extends also to PDSCH data; however, both tools rely on capturing the clear-text registration procedure to address DCI decoding challenges, a condition not always available in typical network scenarios. Overall, prior work addresses the problem only partially and exhibits notable limitations. In particular, 5GSniffer, the only publicly available baseline that does not rely on clear-text registration exchanges, is restricted to PDCCH decoding, while Sni5Gect can access the PDSCH only when registration-phase signaling is observable in clear text. In contrast, our work overcomes these shortcomings by enabling truly blind single-layer PDSCH decoding, without requiring any prior exposure to unencrypted control or registration information. Extending the proposed approach to support multi-layer transmissions represents a natural direction for future work.

3 5G NR Sniffer Challenges

This section outlines key challenges in blind monitoring of 5G NR DL control and data traffic, focusing on decoding Physical Downlink Control Channel (PDCCH) Downlink Control Information (DCI) messages, which are critical for Physical Downlink Shared Channel (PDSCH) decoding. With no prior knowledge, a blind sniffer must simultaneously guess six PDCCH/DCI parameters: the Control Resource Set (CORESET) configuration $C_C = 2^{\lfloor N_{\text{RB}}/6 \rfloor}$, the Aggregation Level (AL) ($C_A = 5$), DCI duration ($C_D = 3$) and DCI size ($C_L \simeq 100$), the 16-bit RNTI ($C_R = 2^{16}$), and the 16-bit scrambling ID ($C_S = 2^{16}$). This already yields a billion-scale hypothesis set of $C_C C_A C_D C_L C_R C_S \approx 2^{58}$ candidate DCIs. Regarding complexity, we report *time* as this sniffer parameter space (hypotheses/validations) and *space* as additional state; $O(1)$ (post caching/pruning) is w.r.t. the corresponding brute-force spaces. NR-internal dimensions are bounded by the standard. After a plausible DCI is found, PDSCH decoding still depends on the user-specific Bandwidth Part (BWP) ($C_B \simeq 10^4$) and on the DM-RS symbol index ($C_M = 14 - \textit{DCI-duration}$), which contribute about 17 bits. This results in billions of



(a) (b)

Figure 1: Comparison of DL resource allocation in a 5G NR cell: (a) single-UE scenario with straightforward BWP establishment; (b) multi-UE scenario where individual BWPs are more challenging to distinguish.

potential tests, which our solutions reduce to $O(1)$ through iterative pruning and caching, once the right parameters are learned. The gNodeB (gNB) may also choose among $C_X \simeq 2^{16}$ NZP-CSI-RS and $C_Z \simeq 2^{12}$ ZP-CSI-RS patterns; these govern how many search spaces coexist in a cell, not the per-UE complexity. We categorize the challenges into seven areas:

[C1] BWP Configuration Inference

5G New Radio (NR) introduces BWPs, which are contiguous subsets of Resource Blocks (RBs) within the total carrier bandwidth (BW). BWPs enable User Equipments (UEs) to operate only over specific portions of the BW, improving spectral efficiency, reducing power consumption, and accommodating diverse service requirements. Each UE can be configured with up to four BWPs per direction (downlink (DL) and uplink (UL)), but only one is active at a time. BWP configurations, including starting RB, size, Subcarrier Spacing (SCS), cyclic prefix type, and frequency location, are securely communicated to the UE via encrypted Radio Resource Control (RRC) signaling. Consequently, sniffers must blindly infer these configurations to accurately interpret resource assignments in DCIs, increasing both complexity and computational requirements. Fig. 1a shows an example of resource allocations within a private 5G cell using srsRAN [41]. Accurately identifying and reconstructing the resources used by each UE is particularly challenging for sniffers in complex multi-UE scenarios (Fig. 1b). This complexity arises because DCI messages on the PDCCH specify allocations relative to the active BWP, encoding information such as data channel resource allocation (PDSCH and Physical Uplink Shared Channel (PUSCH)), Modulation and Coding Scheme (MCS), and essential Hybrid ARQ (HARQ) parameters. Without knowledge of the active BWP, the sniffer cannot reliably map these assignments to the correct resource locations, thus hindering effective control and data channel decoding.

Complexity. *Time:* $O(C_B)$ BWP hypotheses per UE in the worst case (e.g., with ~100 RBs, $C_B \simeq 10^4 \approx 2^{13}$). *Space:* $O(1)$ additional state (enumeration only).

[C2] PDCCH Search Space Localization

In 5G NR, the PDCCH conveys scheduling information through DCI messages. Its configuration includes the CORESET, which defines time–frequency resources, duration, and available Control Channel Elements (CCEs), as well as the AL, specifying the number of aggregated CCEs and, indirectly, the number of occupied Resource Elements (REs). The UE monitors PDCCH search spaces within the CORESET to locate its DCIs. PDCCH search spaces may be *common* (shared among all UEs) or *user-specific*, with periodicity and offset determining monitoring occasions for PDCCH transmissions, and ALs specifying which levels to monitor within the PDCCH search space.

A passive sniffer faces substantial challenges due to the encryption of PDCCH search space parameters (CORESET configurations, ALs, monitoring periodicities), communicated via higher-layer RRC signaling. As a result, the sniffer is forced to perform an exhaustive search for DCI messages across all potential combinations (3GPP standard constrains frequency masks, i.e., 6-RB granularity within the BWP, 1–3 symbol durations, monitoring occasions, and ALs [42]), which is computationally intensive and prone to missing critical DCI messages necessary for monitoring transmissions.

Complexity. *Time:* exhaustive search requires $O(C_C \cdot C_A \cdot C_D)$ hypotheses per UE, where $C_C = 2^{\lfloor N_{RB}/6 \rfloor}$ CORESET frequency bitmaps, $C_A = 5$ ALs, and $C_D = 3$ durations are tested (e.g., $\approx 2^{20}$ trials at ~100 RBs). *Space:* $O(1)$ additional state (bitmaps can be generated on-the-fly).

[C3] DCI Format Ambiguity Resolution

DCI messages inform UEs about resource allocations for both DL PDSCH and UL PUSCH, as well as MCS, HARQ feedback, and power control commands. DCIs vary in size and format, depending on factors such as DL/UL assignments, BWP size, number of configured antennas, and UE capabilities.

DCIs are categorized into *Fallback* formats, simpler, fixed-size formats such as DL 1_0 and UL 0_0 for basic operations, and more flexible *Non-Fallback* formats, DL 1_1, UL 0_1, and formats 2_x supporting advanced NR features, including slot format indication, pre-emption notifications, and UL power control. Registered UEs use RRC conveyed configuration information to calculate the expected DCI length and format. However, a sniffer, lacking access to these encrypted configurations, must attempt decoding considering multiple DCI candidate formats and lengths. This ambiguity significantly increases computational complexity and raises the risk of misinterpreting control information, resulting in failed decoding attempts and ineffective traffic monitoring.

Complexity. *Time:* $O(C_L)$ candidate DCI lengths must be tried per DCI attempt when sizes are unknown (here $C_L \simeq 100 \approx 2^7$). *Space:* $O(1)$ extra state (length enumeration only).

[C4/C5] PDCCH/PDSCH DM-RS & Scrambling

Both PDCCH and PDSCH channels utilize Demodulation Reference Signals (DM-RSs) for precise channel estimation and reliable demodulation of control and data information. These user-specific signals' configurations include the number of symbols, scrambling initialization, and mapping type (A or B), which affect the DM-RSs placement within the slot. DM-RS sequences are generated through a pseudo-random scrambling process initialized by encrypted configuration parameters provided via RRC. Specifically, PDCCH DM-RS sequences are generated using a Scrambling ID (e.g., *pdcch-DMRS-ScramblingID*) combined with the slot number, whereas the PDSCH DM-RSs sequences rely on user-specific scrambling IDs (*scramblingID0*, *scramblingID1*) and the UE's Radio Network Temporary Identifier (RNTI). Since these parameters are not accessible to the sniffer, accurately reconstructing DM-RSs becomes extremely challenging, leading to degraded channel estimation and failures in decoding both PDCCH and PDSCH channels.

Complexity. *Time:* brute-force DM-RS reconstruction is dominated by Scrambling ID enumeration, i.e., $O(C_M \cdot C_S) = O(C_S)$ trials per DM-RS attempt, since C_M (DM-RS symbol-position ambiguity) is a small bounded set and $C_S = 2^{16}$. *Space:* $O(1)$ additional state.

[C6] RNTI and Scrambling ID Extraction

Control and data channels in 5G NR are scrambled using sequences derived from higher-layer signaled parameters, i.e., the DM-RS Scrambling ID, the slot number, and the UE's unique RNTI, effectively randomizing the transmitted signals to mitigate interference and enhance security. Different RNTIs (e.g., C-RNTI, SI-RNTI) may be used depending on the data type. The RNTI specifically scrambles the Cyclic Redundancy Check (CRC) bits appended to the DCI payload, making it essential for decoding control messages. Without knowing the correct RNTI, a sniffer cannot validate decoded DCI, resulting in errors or missed detections. Since both the RNTI and associated Scrambling IDs are communicated through encrypted higher-layer RRC signaling, sniffers must indirectly infer these values. Even if brute-force search to recover these parameters is theoretically possible [4, 5], it incurs substantial computational complexity and may be impractical in multi-UE real-time environments.

Complexity. *Time:* joint brute-force requires $O(C_R \cdot C_S) = 2^{32}$ (RNTI, scrambling ID) pairs to be tested per validation attempt, with $C_R = C_S = 2^{16}$. *Space:* $O(1)$ additional state.

[C7/C8] CSI-RS Configuration Inference

Channel State Information Reference Signals (CSI-RSs) enable UEs to measure channel quality for link adaptation, beam management, and mobility. They include Non-Zero-Power CSI-RS (NZP-CSI-RS), used for channel estimation, and Zero-Power CSI-RS (ZP-CSI-RS), which reserve zero-power REs for interference measurement. Optimizing resource utilization, NZP-CSI-RS and ZP-CSI-RS are typically common to all UEs, and allow the estimation of channel rank for Multiple-input-multiple-output (MIMO) and measurement of interference levels. CSI-RSs configurations, such as resource mapping, density patterns, transmission periodicity, and scrambling IDs, are conveyed via encrypted RRC messages. Their presence affects PDSCH resource allocation, as reserved elements cannot carry data. Since the sniffer lacks access to these configurations, it may misinterpret reserved REs as data-carrying. This leads to errors in demapping, rate matching, and decoding, particularly with ZP-CSI-RS, where distinguishing intentionally blank elements from unscheduled data is challenging. As a result, the risk of incorrect PDSCH decoding and failed CRC checks increases.

Complexity. *Time:* a configuration-enumeration approach requires $O(C_X + C_Z)$ hypotheses to identify NZP-CSI-RS and ZP-CSI-RS parameters (here $C_X \simeq 2^{16}$ and $C_Z \simeq 2^{12}$, i.e., ~16 bits total). *Space:* $O(1)$ additional state.

4 Golden Sniffer: Approach

To overcome the challenges of passive 5G NR monitoring, our approach reduces a naïve sniffer search space of 2^{58} configurations—arising from possible CORESET settings, 16-bit RNTI, 16-bit Scrambling ID, five ALs, three DCI durations, and up to 100 DCI sizes—to at most 50 candidates

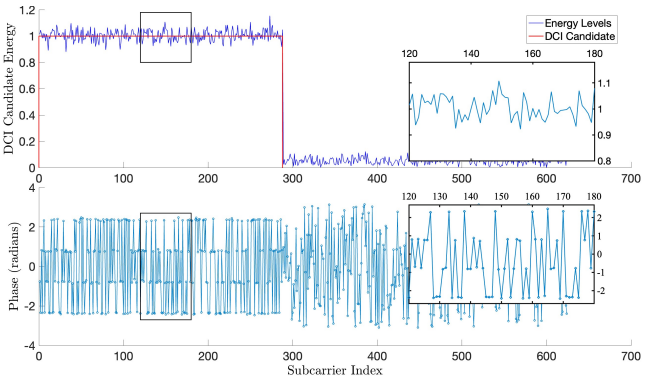


Figure 2: DCI candidates: (top) power levels with red-highlighted candidates (288 carriers); (bottom) IQ symbol phases revealing four distinct phases.

corresponding to *active* DCI sizes. This reduction follows the initial optimized inference of key parameters, which are cached and reused across subsequent decoding operations, ensuring sustained efficiency without repeated exhaustive searches. We detail each solution below.

4.1 Iterative Discovery Techniques

Challenges **[C1]**, **[C2]**, and **[C3]** involve unknown parameters drawn from finite sets and conveyed via encrypted RRC messages. Power distribution analysis substantially reduces the sniffer parameter space by identifying energy patterns across RBs, enabling localization of candidate DCI positions **[S2]** and avoiding exhaustive brute-force searches. We therefore test only compatible ALs and DCI durations, proceeding from larger to smaller allocations to further reduce the search burden. Once a PDCCH message and its associated PDSCH are decoded, the corresponding BWP configuration for that RNTI is cached for reuse.

[S1] Inference of the Active Bandwidth Part

To determine the UE's BWP size and position, *Golden Sniffer* interprets the Frequency Domain Allocation (FDA) in a decoded DCI and confirms it through successful PDSCH decoding. For each feasible BWP size on the carrier [43], the Resource Indication Value (RIV) in a Type 1 FDA is mapped relative to that BWP to a single start-RB and length hypothesis, under which PDSCH decoding is attempted. A Type 0 FDA is a bitmap and its interpretation is straightforward [44]. If the FDA mapping yields a unique BWP configuration and the PDSCH CRC check passes, the UE BWP is cached.

Complexity. *Time:* worst case $O(C_B)$ BWP hypotheses ($C_B \simeq 10^4$) per RNTI; FDA-based pruning restricts trials to the subset consistent with the allocation (strictly fewer than C_B). *Steady-state:* after per-RNTI BWP caching, $O(1)$ hypotheses w.r.t. C_B. *Space:* $O(U)$ cached BWPs (U: # tracked RNTIs).

[S2] PDCCH Localization via Power Clustering

To solve the UE PDCCH search spaces challenge, we directly try to locate the DCI candidates in the first slot symbol, bypassing per-UE CORESET configurations enumeration. To reduce the complexity, we analyze the DCI candidate's power levels, illustrated in Fig. 2, which shows how a candidate will correspond to higher power intensity. We assume non-interleaved PDCCH with contiguous frequency resources, enabling us to restrict the positions of candidate DCIs only to K sharp power clusters, without exhaustive search. Interleaved mapping and non-contiguous frequency masks are left to future work. Fig. 2 also reveals that Quadrature Phase Shift Keying (QPSK) DCI candidate phases assume only four values, hinting at the key intuition behind blind decoding.

Complexity. *Time:* replaces bitmap enumeration over $O(C_C)$ CORESET placements (e.g., $N_{RB} \approx 100 \Rightarrow C_C = 2^{\lfloor N_{RB}/6 \rfloor} \approx 2^{16}$, hence $C_C \cdot C_A \cdot C_D \approx 2^{20}$ hypotheses/slot) with an energy-threshold scan over $O(K)$ clusters (typically $K \leq N_{RB}/6 \approx 16$), i.e., $O(K \cdot C_A \cdot C_D) \lesssim 240 \approx 2^8$ hypotheses/slot. *Steady-state:* caching makes per-cluster $O(1)$, so $T_{\text{slot}} = O(K)$. *Space:* $O(1)$.

[S3] Inference of DCI Formats and Sizes

Inferring the DCI format and size without prior knowledge is challenging due to the variability in DCI lengths. Our approach determines DCI contents through iterative interpretation attempts. While we can successfully identify both *Fallback* (1_0 and 0_0) and *Non-Fallback* (1_1 and 0_1) DCIs, we currently fully interpret only *Fallback* formats for PDSCH decoding; *Non-Fallback* interpretation is left to future work.

Complexity. *Time:* blind decoding tests $O(C_L)$ candidate lengths ($C_L \approx 2^7$), pruning reduces this to at most ≈ 50 candidates, and after caching only the few observed sizes are retried, yielding $O(1)$ w.r.t. C_L. *Space:* worst case $O(C_L)$ (standard-bounded), typically $O(1)$ cached observed-size list.

4.2 DM-RS Sequence Reconstruction

Challenges **[C4]** and **[C5]** in reconstructing PDCCH and PDSCH DM-RSs and scrambling sequences are addressed by exploiting the non-cryptographic nature of 5G NR Gold sequences. By capturing transmitted DM-RS sequences, and applying our proposed technique, we can recover the sequences' initialization parameters. While detailed for PDCCH **[C4]**, the approach also applies to PDSCH **[C5]**, differing mainly in DM-RS symbol mapping and sequence parameters. For PDSCH, we perform a blind search across all Orthogonal Frequency Division Multiplexing (OFDM) symbols in the slot to identify the DM-RS positions and accurately interpret the slot format. However, assuming high-Signal-to-noise Ratio (SNR) regime, QPSK-modulated DM-RS symbols are usually easy to distinguish from the higher-order modulated PDSCH symbols. Thus, by applying our technique, we

can reconstruct the DM-RS sequences for both channels, enabling accurate channel estimation and decoding without prior knowledge of higher-layer configurations.

Gold Sequence Generation and Analysis. The DM-RS sequences are generated using a pseudo-random length-31 Gold sequence $c(n)$ [42]. Our method involves capturing the transmitted DM-RS symbols and estimating the sequence $c(n)$, to obtain initialization parameters, such as scrambling IDs and slot numbers. The sequence $c(n)$ is defined as:

$$c(n) = x_1(n + N_C) \oplus x_2(n + N_C) \tag{1}$$

with $N_C = 1600$. The independent sequences $x_1(n)$ and $x_2(n)$ are defined by the following recursions:

$$\begin{aligned} x_1(n + 31) &= x_1(n + 3) \oplus x_1(n), \\ x_2(n + 31) &= x_2(n + 3) \oplus x_2(n + 2) \oplus x_2(n + 1) \oplus x_2(n) \end{aligned} \tag{2}$$

where $x_1(n)$ is initialized with $x_1(0) = 1$ and $x_1(n) = 0$ for $n = 1, \ldots, 30$, while $x_2(n)$ initialization is given by $c_{\text{init}} = \sum_{i=0}^{30} x_2(i) \cdot 2^i$, and encapsulates several higher-layer parameters of interest depending on sequence's application. When used for the PDCCH DM-RS, c_{init} includes the slot number, the symbol index, and the Scrambling ID, which is securely delivered only to the intended user. Since $x_1(n)$ initialization is known, the sequence $x_1(n + N_C)$ can be precomputed. Our goal is to estimate c_{init} from a sequence of observations $\hat{c}(n)$. Given that $x_1(n)$ is known, the estimated sequence $\hat{c}(n)$ can be associated with an estimate of $x_2(n)$ as:

$$\hat{x_2}(n + N_C) = \hat{c}(n) \oplus x_1(n + N_C) \tag{3}$$

where $x_1(n + N_C)$ is precomputed. The recursion for $x_2(n)$ can then be reformulated in terms of future values:

$$x_2(n) = x_2(n + 1) \oplus x_2(n + 2) \oplus x_2(n + 3) \oplus x_2(n + 31) \tag{4}$$

allowing for the estimation of the initialization vector c_{init} from any set of 31 consecutive values of the sequence $c(n)$.

The sequence $c(n)$ is used to generate the DM-RS sequences $r_l(m)$. Indeed, for the l-th OFDM symbol, $r_l(m)$ is defined by the QPSK mapping:

$$\sqrt{2} r_l(m) = (1 - 2 \cdot c(2m)) + j(1 - 2 \cdot c(2m + 1)) \tag{5}$$

Therefore, to obtain 31 consecutive values of $c(n)$, we need only 16 consecutive DM-RS symbols $r_l(m)$.

Mapping of DM-RS in PDCCH. The DM-RS sequence $r_l(m)$ is mapped to physical resources as:

$$a_{k,l}^{(p,\mu)} = \beta_{\text{DMRS}}^{\text{PDCCH}} \cdot r_l(3n + k') \tag{6}$$

where $a_{k,l}^{(p,\mu)}$ is the transmitted symbol at subcarrier k and OFDM symbol l, for antenna port p and numerology μ, $\beta_{\text{DMRS}}^{\text{PDCCH}}$ is the power scaling factor for the PDCCH DM-RS, $k = nN_{\text{sc}}^{\text{RB}} + 4k' + 1$ with $k' = 0, 1, 2$, and $n \geq 0$.

Since the number of subcarriers in a RB $N_{\text{sc}}^{\text{RB}}$, is equal to 12, we obtain $k = 4(3n + k') + 1$, where $n' = 3n + k'$ can take every possible integer value. Consequently, Eq. (6) becomes:

$$a_{4n'+1,l}^{(p,\mu)} = \beta_{\text{DMRS}}^{\text{PDCCH}} \cdot r_l(n'). \tag{7}$$

Hence, the DM-RS symbols are transmitted every fourth subcarrier, starting at index 1 within the PDCCH RBs.

Phase Analysis for Symbol Estimation. Our strategy exploits the limited alphabet of the QPSK mapping of $c(n)$ to infer the full PDCCH message and its embedded DM-RS sequence by analyzing the estimated phase values at the receiver. This is shown in Fig. 2 (bottom), which displays the four distinct phase states of DCI candidate regions. Let $\Delta\Phi_{k,l}$ denote the $\pi/2$-spaced phase increment estimate with minimum angular distance from the observed phase increment:

$$\Delta\Phi_{k,l} = \arg \min_{\Phi \in \{0, \pm\pi/2, \pi\}} \left| e^{j(\angle y_{k+1,l} - \angle y_{k,l})} - e^{j\Phi} \right| \tag{8}$$

which, under Line-of-sight (LOS)-dominant path and high-SNR regime assumptions, equals $\Delta\Phi_{k,l} = \angle a_{k+1,l} - \angle a_{k,l}$ (see App. A for the derivation), enabling candidate symbol sequences reconstruction:

$$\hat{a}_{k_0+\Delta k,l} = \hat{a}_{k_0,l} e^{j \sum_{k=k_0}^{k_0+\Delta k-1} \Delta\Phi_{k,l}}, \tag{9}$$

where the four possibilities stem from the initial choice of $\hat{a}_{k_0,l}$. Each reconstructed sequence embeds a candidate DM-RS, obtained from (7) as $\hat{r}_l(n') = \hat{a}_{4n'+1,l}$, which, in turn, yields the following tentative Gold sequence:

$$\hat{c}(n) = \begin{cases} \text{Re}\{\hat{r}_l(n)\} < 0, & n \text{ even} \\ \text{Im}\{\hat{r}_l(n)\} < 0, & n \text{ odd} \end{cases}$$

Since the sequence $x_1(n)$ is known, the candidate sequence $\hat{c}(n)$ corresponds to the one for which the estimated sequences $\hat{x_2}(n)$, as in (3), satisfies the recursive formula in (2). Given a suitable candidate, we can compute N_C more terms in order to recover the initial state c_{init}.

Detailed OFDM signal model, channel and assumptions are provided in App. A. The QPSK mapping of $c(n)$ for 5G NR reference signals is defined in [42]. App. B summarizes the four key cases relevant to this work (PBCH, PDCCH, single-layer PDSCH, and NZP-CSI-RS DM-RS).

[S4]/[S5] DM-RS Recovery & Scrambling Estimation

After estimating the initialization value c_{init}, we solve for the Gold sequence generation parameters, such as the Scrambling ID N_{ID}, the slot number $n_{\text{s,f}}^{\mu}$, and the OFDM symbol index l. For the PDCCH DM-RS, c_{init} is defined as [42]:

$$c_{\text{init}} = [2^{17} \cdot x \cdot (2N_{\text{ID}} + 1) + 2N_{\text{ID}}] \bmod 2^{31} \tag{10}$$

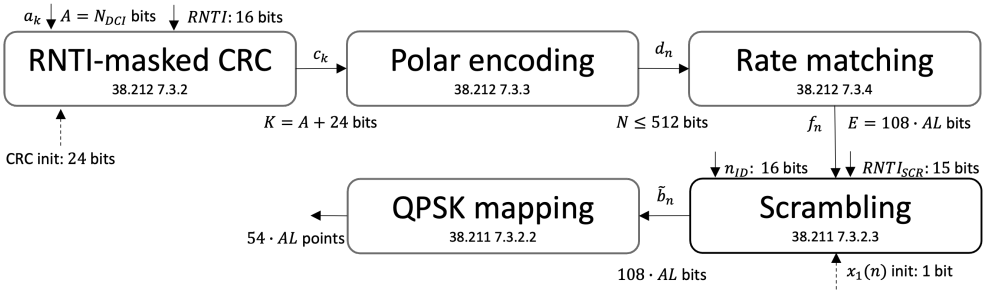


Figure 3: PDCCH DCI message generation steps.

where $x = N_{\text{symb}}^{\text{slot}} \cdot n_{\text{s,f}}^{\mu} + l + 1$. Here, $N_{\text{symb}}^{\text{slot}}$ is the number of OFDM symbols per slot (typically 14), $n_{\text{s,f}}^{\mu}$ is the slot number within a frame (with $0 \leq n_{\text{s,f}}^{\mu} < 10 \cdot 2^{\mu}$), and l is the OFDM symbol index within the slot ($0 \leq l < N_{\text{symb}}^{\text{slot}}$). The parameter N_{ID} is either provided as the higher-layer parameter *pdcch-DMRS-ScramblingID* or, if not available, the Physical Cell ID (PCI) $N_{\text{ID}}^{\text{cell}}$. Given c_{init}, we extract 16-bit N_{ID} as $(c_{\text{init}}/2) \bmod 2^{16}$, and we obtain x from (10) checking that $x \leq 140 \cdot 2^{\mu}$ matches the slot and symbol index.

Complexity. *Time:* recovers c_{init} from DM-RS, avoiding $O(C_S)$ Scrambling ID enumeration ($C_S = 2^{16}$) and requiring $O(1)$ candidate reconstructions w.r.t. C_S (and bounded C_M). Residual slot/symbol ambiguity is finite and becomes $O(1)$ after caching. *Space:* $O(1)$ (plus cached slot/symbol).

4.3 RNTI and Scrambling ID Inference

Once the DM-RS initialization and the corresponding Scrambling ID are determined, we recover the user-specific RNTI and Scrambling ID **[C6]** by exploiting the linearity of the PDCCH encoding and scrambling operations. Specifically, the scrambled bit sequence $\tilde{b}(i) = b(i) \oplus c(i)$ exhibits a linear dependence (modulo 2) on unknown quantities (DCI payload, RNTI, and Scrambling ID n_{ID}), as well as on known parameters (CRC and $x_1(n)$ sequence initialization). As the encoded sequence $b(i)$ is not random but highly structured due to the use of linear block codes and CRCs, these linear dependencies are preserved under scrambling and can be exploited to jointly infer the RNTI and other unknown parameters. Unlike exhaustive brute-force approaches requiring up to 2^{31} trials, our method formulates the problem as a linear inversion that operates in constant time $O(1)$ once the set of feasible DCI sizes is known, drastically reducing computational complexity.

Fig. 3 illustrates the DCI generation pipeline, in which each processing stage is linear over the binary field $\mathbb{F}_2$.

Step 1: CRC Attachment and RNTI Masking. The DCI payload $\mathbf{a}$ consists of $A = N_{\text{DCI}}$ bits. A 24-bit CRC is appended to $\mathbf{a}$ to form the sequence $\mathbf{c}$ of length $K = A + 24$ bits. This combination of *CRC Computation and Attachment* with *RNTI Masking* can be expressed as:

$$\mathbf{c} = \underbrace{\begin{bmatrix}\mathbf{a}_{\text{init}}^{\text{CRC}} & \mathbf{a}\end{bmatrix}\begin{bmatrix}\mathbf{0}_{24\times A} & \mathbf{G}_{K\times 24}^{\text{CRC}} \\ \mathbf{I}_A\end{bmatrix}}_{\text{CRC Computation and Attachment}} + \underbrace{\mathbf{a}_{\text{RNTI}}\begin{bmatrix}\mathbf{0}_{16\times K-16} & \mathbf{I}_{16}\end{bmatrix}}_{\text{RNTI Masking}} \quad (11)$$

In *CRC Computation and Attachment*, $\mathbf{a}_{\text{init}}^{\text{CRC}}$ is the 24-bit initial CRC register state, and $\mathbf{G}_{K\times 24}^{\text{CRC}}$ is the CRC generator matrix that transforms the payload $\mathbf{a}$ into the appended CRC bits. The $A \times A$ identity matrix $\mathbf{I}_A$ preserves the original unaltered payload $\mathbf{a}$ within $\mathbf{c}$, while the zero matrix $\mathbf{0}_{24\times A}$ hides the initialization from the output.

The CRC computation fundamentally calculates the remainder of a division by a generator polynomial, a linear operation over the binary field $\mathbb{F}_2$. Note that the CRC register is initialized to a non-zero value so that the operation results non-linear, i.e., affine linear. Embedding this initialization in the unknowns renders the overall operation linear by decomposing the CRC output into two linear components: Free Evolution (i.e., the output from initial CRC state with no input, $\mathbf{a} = \mathbf{0}$) and Forced Evolution (i.e., the output from a zero initial CRC state with actual input payload $\mathbf{a}$). Thus, the CRC output can be expressed as the sum in $\mathbb{F}_2$ of these two linear contributions. In *RNTI masking*, the 16-bit RNTI $\mathbf{a}_{\text{RNTI}}$ is XORed with the CRC, uniquely binding the control message to its user. The zero matrix $\mathbf{0}_{16\times K-16}$ prevents RNTI influence on payload bits, while the identity matrix $\mathbf{I}_{16}$ aligns RNTI bits with the CRC portion of $\mathbf{c}$. The matrix formulation in Eq. (11) demonstrates that $\mathbf{c}$ is a linear combination of $\mathbf{a}$ and the RNTI, ensuring that both CRC attachment and RNTI masking are linear operations over $\mathbb{F}_2$.

Step 2: Polar Encoding and Rate Matching. The sequence $\mathbf{c}$ is polar encoded using a generator matrix $\mathbf{G}_{\text{polar}} \in \mathbb{B}^{K\times N}$ to produce a N-bit codeword $\mathbf{d} = \mathbf{c} \cdot \mathbf{G}_{\text{polar}} \in \mathbb{B}^{N}$. Subsequently, $\mathbf{d}$ undergoes rate matching via matrix $\mathbf{G}_{\text{RM}}$ to fit the allocated PDCCH resources, yielding a sequence $\mathbf{f} = \mathbf{d} \cdot \mathbf{G}_{\text{RM}}$. with length $E = 108 \times \text{AL}$, i.e., the number of CCEs assigned. This linear transformation selects, punctures (if $E < N$), or repeats bits (if $E > N$) from $\mathbf{d}$ to achieve the required code rate. Since each CCE spans $N_{\text{sc}}^{\text{CCE}} = 6 \times N_{\text{sc}}^{\text{RB}} = 72$ subcarriers, and one out of every four allocated subcarriers is reserved for DM-RS,

only the remaining three-fourths, i.e. $54 \times$ AL subcarriers are available for DCI data. Given that QPSK modulation maps each subcarrier to 2 bits, the total bits that can be transmitted for the DCI is $E = 108 \times$ AL bits, ensuring that the rate-matched sequence $\mathbf{f}$ fits within the allocated PDCCH resources, according to the AL and modulation scheme.

Step 3: Scrambling. The rate-matched sequence $\mathbf{f}$ is scrambled by XORing it with a scrambling sequence $\mathbf{c}_{\text{scr}}$ generated from $c(n)$ as $\tilde{\mathbf{b}} = \mathbf{f} \oplus \mathbf{c}_{\text{scr}}$, which depends on the RNTI RNTI_{SCR} and the Scrambling ID n_{ID}. The scrambling operation is linear, and $\mathbf{c}_{\text{scr}}$ can be represented as:

$$\mathbf{c}_{\text{scr}} = [\mathbf{f}, \mathbf{a}_{\text{RNTI}_{\text{SCR}}}, \mathbf{a}_{n_{\text{ID}}}, \mathbf{a}_{\text{init}}^{x_1}] \cdot \begin{bmatrix} \mathbf{I}_E \\ \mathbf{G}_{E\times 31}^{\text{SCR}} \\ \mathbf{G}_{E\times 1}^{x_1} \end{bmatrix} \tag{12}$$

where $\mathbf{G}_{E\times 31}^{\text{SCR}}$ and $\mathbf{G}_{E\times 1}^{x_1}$ represent the relationship between the scrambling sequence and its initializations.

Step 4: Modulation and Mapping. The scrambled E bits $\tilde{\mathbf{b}}$ are mapped to QPSK symbols and transmitted over the allocated PDCCH physical resources. The PDCCH DM-RS subcarriers can be exploited to extract the *pdcch-DMRS-ScramblingID* using the phase-based method in Sec. 4.2.

[S6] Joint RNTI and Scrambling ID Inference

By combining and rearranging the described operations, we can represent the entire process as a single linear equation:

$$\tilde{\mathbf{b}} = [\mathbf{a}, \mathbf{a}_{\text{RNTI}}, \mathbf{a}_{\text{RNTI}_{\text{SCR}}}, \mathbf{a}_{n_{\text{ID}}}, \mathbf{a}_{\text{init}}^{\text{CRC}}, \mathbf{a}_{\text{init}}^{x_1}] \cdot \mathbf{G}_{\text{DCI}} = \mathbf{a}_{\text{tot}} \cdot \mathbf{G}_{\text{DCI}} \tag{13}$$

where $\mathbf{a}_{\text{tot}}$ is the vector of unknowns and known initialization, and $\mathbf{G}_{\text{DCI}}$ is the combined generator matrix representing all encoding and scrambling operations. Given that the initializations $\mathbf{a}_{\text{init}}^{\text{CRC}}$ and $\mathbf{a}_{\text{init}}^{x_1}$ are known, the unknown components can be isolated as follows:

$$\tilde{\mathbf{b}} - [\mathbf{a}_{\text{init}}^{\text{CRC}}, \mathbf{a}_{\text{init}}^{x_1}] \cdot \mathbf{G}_{\text{DCI}}^{\text{known}} = [\mathbf{a}, \mathbf{a}_{\text{RNTI}}, \mathbf{a}_{\text{RNTI}_{\text{SCR}}}, \mathbf{a}_{n_{\text{ID}}}] \cdot \mathbf{G}_{\text{DCI}}^{\text{unk}} \tag{14}$$

A solution is considered valid, according to [42] Clause 7.3.2.3, if either the CRC masking RNTI matches the RNTI estimated from DM-RS and the Scrambling ID matches the DM-RS Scrambling ID, i.e., $\mathbf{a}_{n_{\text{ID}}} = N_{\text{ID}} \wedge \mathbf{a}_{\text{RNTI}} \bmod 2^{15} = \mathbf{a}_{\text{RNTI}_{\text{SCR}}}$ with N_{ID} from (10); or if the Scrambling ID coincides with PCI and the RNTI estimated from the PDCCH scrambling is zero, i.e., $\mathbf{a}_{n_{\text{ID}}} = N_{\text{ID}}^{\text{cell}} \wedge \mathbf{a}_{\text{RNTI}_{\text{SCR}}} = 0$.

This system comprises $A + 47$ unknowns, including the A-bit DCI payload, the 16-bit RNTI, the possible 15-bit RNTI_{SCR}, and the 16-bit Scrambling ID n_{ID}, and $108 \times$ AL equations (the length of $\tilde{\mathbf{b}}$). When $108 \times \text{AL} > A + 47$, the system is overdetermined and yields a unique solution. However, as A increases, the risk of false detections increases; with this approach, for $A > 50$, the sniffer requires AL > 1. Although $A = N_{\text{DCI}}$ is initially unknown, once a valid solution is found, it can be cached to streamline decoding of subsequent RNTI-related DCIs, reducing computational complexity.

Complexity. *Time:* replaces (C_R, C_S) brute-force (i.e., $C_R C_S = 2^{32}$ RNTI/scrambling ID pairs) with one solve-and-validate per candidate DCI size: $O(C_L)$ solves per DCI attempt during initial size discovery; after caching, only the few observed sizes are retried, yielding $O(1)$ w.r.t. C_L. Per-size validation is $O(1)$ w.r.t. (C_R, C_S) (no 2^{32} scan). *Steady-state:* with per-RNTI cached parameters, $O(1)$ w.r.t. (C_R, C_S, C_L). *Space:* $O(U)$ per-RNTI cache (scrambling IDs, size).

4.4 Detection of CSI-RS

After recovering user-specific control parameters and enabling blind PDSCH decoding, a remaining source of decoding failures arises from the presence of CSI-RS. These reference signals reserve specific REs within the DL grid for channel sounding and interference measurements, and are configured via encrypted higher-layer signaling. A passive sniffer that is unaware of their placement may incorrectly interpret such reserved resources as data-carrying, leading to systematic demapping and CRC failures.

In this stage, we therefore address the blind identification of both NZP-CSI-RS and ZP-CSI-RS. While the former can be directly detected through their transmitted symbols, the latter require an indirect inference strategy, as they intentionally occupy resources with zero power.

[S7] NZP-CSI-RS Detection

Detection of single-layer NZP-CSI-RS can be done using the phase-based method detailed in Sec. 4.2 and App. B. Extension to multi-layer NZP-CSI-RS is left for future work.

Complexity. *Time:* avoids $O(C_X)$ enumeration ($C_X \simeq 2^{16}$) via phase-based detection, yielding $O(1)$ hypotheses w.r.t. C_X after caching. *Space:* $O(1)$ per cached NZP-CSI-RS resource.

[S8] Adaptive ZP-CSI-RS Detection and Exclusion

Conversely, to infer the presence of ZP-CSI-RS, we adopt an adaptive method that leverages unsuccessful PDSCH decoding attempts to guide inference. Following a failed PDSCH decode, we perform a per-slot power analysis and locate REs patterns with consistently low power (zero in the ideal noiseless case), which we interpret as candidate ZP-CSI-RS locations. We then adjust the rate-unmatching process to exclude these REs and attempt PDSCH decoding again. If decoding succeeds after adjusting for inferred CSI-RS locations, we cache the CSI-RS configuration for future attempts, enabling the sniffer to track dynamic CSI-RS settings without any prior configuration knowledge.

Complexity. *Time:* replaces $O(C_Z)$ pattern testing ($C_Z \simeq 2^{12}$) with adaptive power-based failure analysis to infer and cache ZP exclusion positions, yielding $O(1)$ w.r.t. C_Z in steady state. Space: $O(1)$ per cached ZP-CSI-RS configuration.

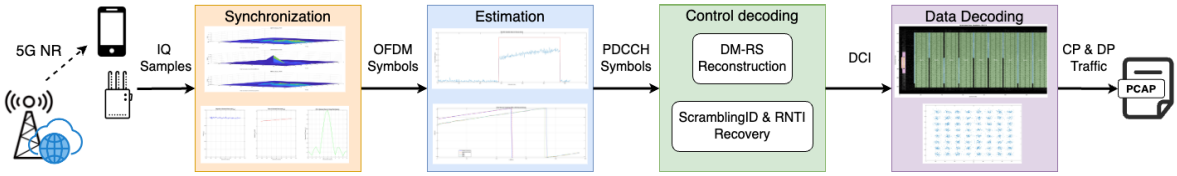


Figure 4: Operational workflow of *Golden Sniffer*, showing the four sniffing main stages.

4.5 Overall Golden Sniffer workflow

We conclude this section by tying together the proposed mechanisms into a coherent operational process. Fig. 4 presents the end-to-end *Golden Sniffer* pipeline, providing a consolidated view of how the previously described techniques are integrated to enable robust decoding of 5G NR DL control and data channels from captured raw IQ samples.

Stage 1: Synchronization, System Information, and NZP-CSI-RS. *Golden Sniffer* (i) processes raw IQ data to synchronize to the cell by detecting the Synchronization Signal Block (SSB) and decoding the Physical Broadcast Channel (PBCH) to obtain basic cell parameters and maintain synchronization; it then (ii) iterates over all feasible NZP-CSI-RS configurations, using the phase-based method to detect their positions and periodicity [S7].

Stage 2: PDCCH Localization and DM-RS Detection. *Golden Sniffer* applies energy thresholding to detect candidate PDCCH allocations [S2], iterates over feasible CORESETs and ALs, and uses the phase-based method to detect PDCCH DM-RS sequences and their initialization parameters [S4].

Stage 3: Control Channel Decoding (PDCCH). *Golden Sniffer* iterates over feasible DCI sizes [S3] and applies linear inversion to jointly recover the user-specific RNTI and Scrambling ID without brute-force search [S6].

Stage 4: Data Channel Decoding (PDSCH). To infer the user-specific BWP configuration [S1], *Golden Sniffer* iterates over feasible interpretations of the frequency-domain allocation, searches for PDSCH DM-RS positions and initialization parameters [S5], regenerates the corresponding Gold sequences for channel estimation, and attempts decoding. Upon successful decoding, the recovered parameters are cached and the decoded data are optionally stored; otherwise, ZP-CSI-RS detection [S8] is applied and decoding is retried.

5 Experimental Assessment

In this section, we evaluate the performance of *Golden Sniffer* in both controlled and real-world 5G NR environments, allowing us to assess robustness, scalability, and practical applicability. We specifically consider the following two diverse network configurations:

- **Controlled Testbed:** A private Software-Defined Radio (SDR)-based 5G SA deployment built on srsRAN [41] and Open5GS [45], using commercial-off-the-shelf (COTS) UEs and configurable BW, SCS, and transmit power. While inherently limited in scale by SDR/RF front-ends and available spectrum, this setup enables systematic, moderate-scale scalability experiments. Importantly, access to gNB logs provides ground truth, allowing us to precisely stress-test *Golden Sniffer* under controlled yet realistic conditions.
- **Commercial Networks:** 5G operators' cells in the n78 band (FSVA3044 VSA [46] at 3680.01 MHz, with 217 RBs for 80 MHz BW, $\mu = 1$, and PCI 710) and in the n28 band (USRP N310 [47] at 763 MHz, with 52 RBs for 10 MHz BW, $\mu = 0$, and PCI 175). These captures span both FDD and TDD modes, confirming that *Golden Sniffer* can decode real-world DL traffic under complex operational configurations, including user-specific BWPs, multi-symbol DCIs, and user-specific scrambling IDs.

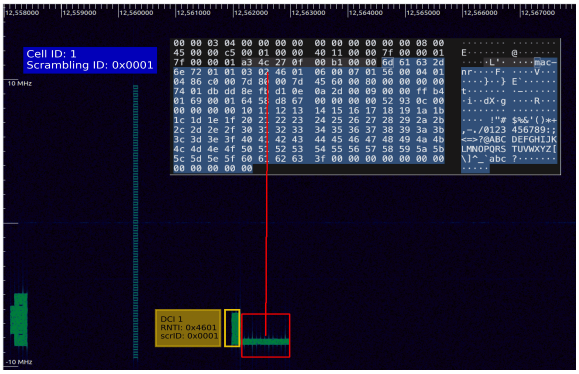


Figure 5: Wireshark capture of an ICMP Echo Reply recovered from PDSCH with encryption disabled.

Concerning **Hardware & Processing**, raw IQ data was captured using a USRP X310 SDR and processed by a *MATLAB*-based prototype running on an Apple M3 Pro CPU. Processing was performed offline and required approximately 30 s per 1 s of captured data. Notably, our current implementation has not yet undergone algorithmic or platform-specific optimization. Despite this, the observed processing time suggests that a future optimized implementation and porting to a compiled language could realistically enable real-time operation on commodity hardware.

5.1 Controlled Testbed: UE Traffic Profiling

To evaluate *Golden Sniffer* for network analytics, we deployed a 5G testbed with UE-specific BWPs and multi-symbol DCIs, where competing approaches [4, 5] are ineffective.

Functional Validation. We disabled encryption in our testbed and validated user-plane decoding by intercepting PDSCH transmissions and recovering ICMP Echo Replies, as shown in Fig. 5. The results confirm correct demodulation and decoding when security is disabled.

Application-Traffic Profiling. When encryption is enabled, user-plane payloads remain undecipherable. However, *Golden Sniffer* can still extract control-plane information from control messages and PDSCH metadata (e.g., MCS, Transport Block Size (TBS), scheduling times). Fig. 6 illustrates time-frequency allocation histograms for two different application scenarios: streaming traffic (blue) and e-commerce (orange). Even though user data was encrypted, these distinct allocation patterns clearly emerge from the decoded PDCCH messages and PDSCH resource mappings. Such patterns can be fed into higher-layer analytics or classification models to infer application behavior. This highlights *Golden Sniffer*'s versatility for network monitoring, diagnostics, and traffic classification, even under fully encrypted conditions.

5.2 Decoding Performance

Leveraging the controlled testbed, we quantify *Golden Sniffer*'s DCI decoding reliability; unless noted, each test averages 20 independent 1 s captures to ensure statistical validity.

AWGN Sensitivity. To quantitatively assess *Golden Sniffer*'s performance, we evaluated its decoding success rate against decreasing PDCCH SNR by injecting Additive White Gaussian Noise (AWGN) into a high-SNR (23.7 dB) IQ capture. Fig. 7 reveals a waterfall curve for both UL and DL DCI grants: a high-SNR plateau with over 90% success above 16.5 dB, a sharp performance cliff where success drops from 88.7% at 15.4 dB to 12.9% at 12.8 dB, and a failure floor with a near-zero success rate below 12.5 dB.

We validated this trend against a suitable theoretical benchmark. Since our method's success is ultimately determined by the integrity of the 63-bit information block (39-bit DCI payload + 24-bit CRC), extended to a transmitted sequence $E = 108 \times 4 = 432$ bits (assuming $AL = 4$), a fitting benchmark is the block success probability over these 432 bits. Derived from the Bit Error Rate (BER) for Gray-coded QPSK $P_b(\gamma_b) = Q(\sqrt{2\gamma_b})$, with γ_b the SNR per bit, the block success probability is $P_{\text{succ}}^{\text{DCI}}(\gamma_b) = \left[1 - P_b(\gamma_b)\right]^{432} = \left[1 - Q(\sqrt{2\gamma_b})\right]^{432}$. The shape of our measured data aligns remarkably well with this theoretical curve after accounting for an estimated 7 dB implementation loss from our SDR-based receiver. This alignment supports the robustness of *Golden Sniffer* approach.

Multi-configuration evaluation. We assessed the impact of duplex mode, number of UEs, and slicing (UE-specific BWPs with varying sizes) on DCI detection rate and decoding time. Across 80 high-SNR captures (>26 dB and >1000 DCIs), we generated heterogeneous traffic profiles including ICMP ping, YouTube streaming, and shopping browsing, yielding diverse MCS and PDSCH sizes. Results in Tab. 2 show slicing increases decoding time by multiplying BWPs and search spaces, while TDD requires more time probably due to fragmented, non-contiguous DL slots adding synchronization and decoding overhead.

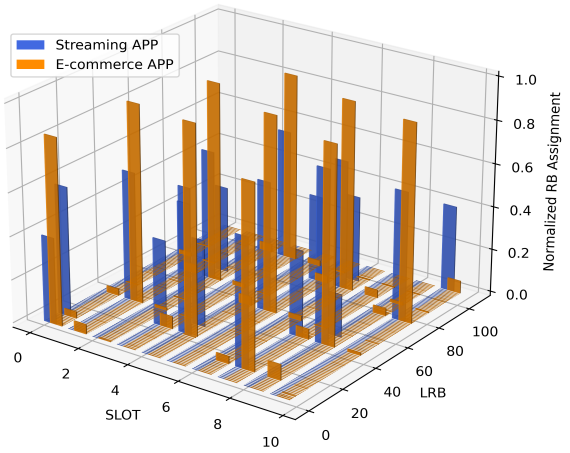


Figure 6: RB assignment histograms over consecutive slots: streaming (blue) vs e-commerce (orange) traffic.

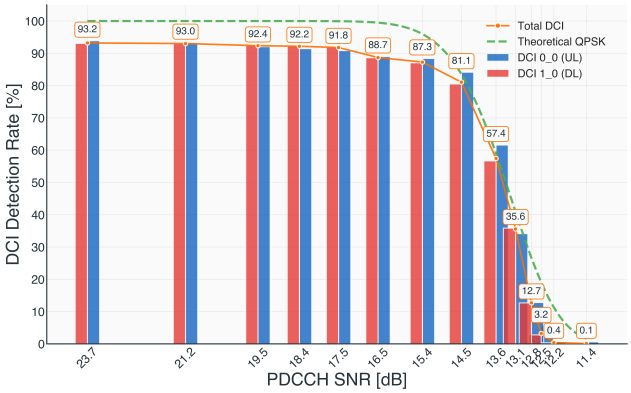


Figure 7: DCI detection rate vs. SNR. Measured data (orange line and bars) is benchmarked against the theoretical block success probability (dashed green line).

SOTA comparison. Compared to 5GSniffer [5], the only publicly available sniffer that does not rely on clear-text registration procedures, *Golden Sniffer* demonstrates substantial improvements in both detection accuracy and runtime efficiency. When evaluated on identical high-SNR IQ captures (Tab. 2), our tool achieves DCI detection rates exceeding 92% in the Single-UE case and 97% in the Multi-UE FDD scenario, whereas 5GSniffer attains detection rates of only 2-3% in the same conditions. Furthermore, in both scenarios, *Golden Sniffer* completes the full decoding pipeline in under 20 s, achieving a runtime improvement of almost four orders of

Table 2: Performance comparison: DCI detection rate and total decoding time across various scenarios.

		Golden Sniffer		5GSniffer	
Scenario	#UE	Detection rate [%]	Time [s]	Detection rate [%]	Time [s $\times 10^3$]
Single-UE FDD	1	92.2 ± 3.1	17.9 ± 1.6	2.2 ± 1.0	123.1 ± 10.2
Multi-UE FDD	2	97.2 ± 2.0	19.2 ± 8.5	3.0 ± 2.1	129.8 ± 11.8
Multi-UE TDD	2	94.7 ± 1.9	50.0 ± 3.2	N/A	N/A
Sliced, FDD (#2, BWPs)	3	98.5 ± 0.5	50.2 ± 1.3	N/A	N/A

* N/A: unsupported configurations.

magnitude with respect to 5GSniffer[1]. This performance gap stems from a fundamental change in approach, replacing exhaustive brute-force exploration with an algorithmic strategy that exploits structural properties of the 5G NR physical layer to sharply reduce the sniffer search space.

NLOS robustness. To evaluate *Golden Sniffer*'s robustness, we assessed its decoding success rate under Non-line-of-sight (NLOS) conditions using 3GPP Clustered Delay Line (CDL) channel models [48]. CDL models simulate realistic multipath via clustered taps with configurable delay spreads and angular dispersion. We conducted 10^4 decoding attempts per scenario, fixing the Doppler shift to 1 Hz (quasi-static) while varying the channel model, propagation environment, and delay spread. In a moderately dispersive outdoor environment (CDL-A, open area, 300 ns delay spread), the sniffer achieved a DCI detection rate of 70 ± 0.46%, indicating good resilience to limited multipath. In an urban macrocell setting (CDL-C, 500 ns), characterized by denser scattering and more reflective surfaces, performance dropped to 26 ± 0.44%. Under a severely dispersive urban NLOS condition (CDL-E, 1 µs), decoding success fell to 12 ± 0.33%, highlighting the challenges posed by rich multipath and long delay spreads.

Timing micro-benchmarks. To profile *Golden Sniffer*, we timed its four workflow stages (see Fig. 4) in the three identified operational SNR zones. In the high-SNR plateau (≥ 19.6 dB), Stage 3 consumed 97.4 ± 1.8% of the runtime, with Stages 1, 2, and 4 taking 0.2 ± 0.09%, 1.4 ± 0.99%, and 1.0 ± 0.74%, respectively. This pattern held on the performance cliff (15.4–12.8 dB), where Stage 3's share increased slightly to 98.0 ± 1.0%, while the other steps accounted for 0.2 ± 0.23%, 1.1 ± 0.72%, and 0.7 ± 0.11%. In the failure floor region below 12.5 dB, the sniffer could not complete decoding. Results confirm that the blind DCI search in Stage 3 is the bottleneck.

[1]Runtimes for 5GSniffer were measured under an experimental cap of 30 tested scrambling IDs and re-normalized to 2^{16}. Beyond this cap, the host encountered an out-of-memory error.

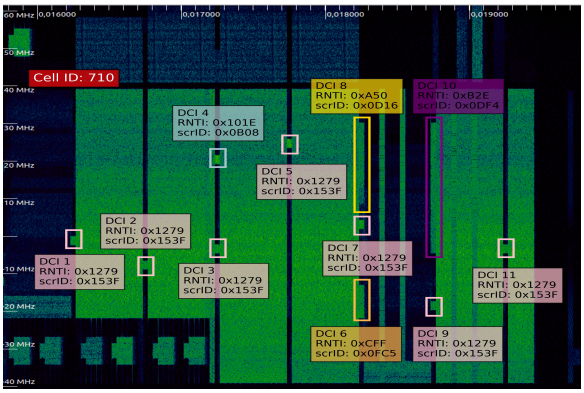


Figure 8: 5G n78 commercial network: highlighted DCI's with user-specific RNTI and Scrambling ID.

5.3 Tests in Operational Networks

To validate *Golden Sniffer* in real-world deployments, we conducted in-the-wild tests on commercial national operator networks. In these scenarios, despite the variability in network configurations and environmental challenges, *Golden Sniffer* consistently extracted unique Scrambling IDs and RNTIs from the DL control channels.

Validation. Our analysis reveals that the common assumption used in competing approaches [4, 5], that the Scrambling ID defaults to the PCI, does not hold in operational networks. Indeed, we observed that commercial operators often configure user-specific scrambling parameters to enhance security, rendering brute-force methods ineffective. Fig. 8 illustrates a commercial capture where multiple *Non-Fallback* DCI formats are observed. User-specific RNTIs and Scrambling IDs are color-coded to clearly demonstrate that *Golden Sniffer* reliably differentiates users and decodes control information, even in complex commercial network.

6 Countermeasure

Our experiments reveal that user traffic can be effectively intercepted. To address this issue, we propose a countermeasure based on strategically inserting zeros within the PDCCH DM-RS symbols in the frequency domain. This, emulating deep fading, disrupts blind DM-RS reconstruction [S4] (Fig. 9), causing attacker PDCCH decoding to fail, whereas legitimate UEs, which know scrambling parameters, still reconstruct the DM-RS and decode with limited performance impact. This can be achieved by modulating the amplitude of the frequency domain PDCCH REs at the gNB.

Validation. We performed both MATLAB simulations and experimental validation with srsRAN and COTS UEs, to analyze the trade-off between security effectiveness and signal quality degradation. For each AL, we identified the optimal number of inserted zeros that render *Golden Sniffer* ineffective (previously achieved a 94% DCI-decoding rate) and

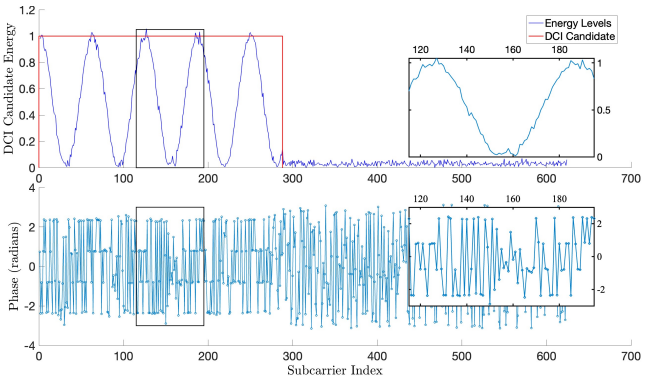


Figure 9: DCI candidates: (top) energy levels with a red-highlighted candidate; (bottom) IQ symbol phases revealing unpredictable phases at zero crossings.

the associated SNR penalty for a 40-bit DCI, yielding the minimal-penalty configurations shown in Tab. 3. The results show that lower ALs incur a higher penalty due to their higher coding rate, making the polar decoder more sensitive to inserted zeros. Nevertheless, legitimate UEs experience a limited impact on channel quality with up to 7 dB SNR degradation, while *Golden Sniffer* becomes ineffective. This embodies a standard security-reliability trade-off and demonstrates the practical feasibility of enhancing privacy without substantial degradation of network performance.

Furthermore, the approach is robust against MIMO attackers. Since the countermeasure targets the PDCCH, which is transmitted isotropically, a multi-antenna attacker receives the same artificially faded signal on all antennas and cannot use spatial processing to undo the effect.

7 Limitations and Future Work

Golden Sniffer demonstrates feasible blind low-complexity 5G NR DL-only traffic analysis, yet several practical barriers remain. First, it presumes to capture the same beam that serves the victim; adaptive gNB beamforming can place deep spatial nulls elsewhere, so interception is possible only when attacker and UE fall inside the same beam. Second, the prototype handles single-layer PDSCH only. A multi-layer allocation would require the adversary to equip at least as many receive antennas as spatial layers, estimate the gNB antenna count from CSI-RS, and infer the per-UE layer number from the DM-RS port set. Even with this information, the attacker's equivalent MIMO channels may differ from those of the UE, complicating layer separation and calling for advanced detection methods. Finally, future work must address line-rate processing, UL interception which typically relies on different sequences, e.g., Zadoff-Chu, *Non-Fallback* DCIs interpretation, and harden the system against beamforming dynamics and high-rank MIMO to remain effective across heterogeneous large-scale commercial 5G scenarios.

Table 3: Minimal-penalty zero-insertion configurations per AL and DCI duration, yielding a 100% sniffer-failure rate while legitimate UEs keep decoding.

AL	DCI Duration 1		DCI Duration 2		DCI Duration 3	
	# Zeros	Penalty [dB]	# Zeros	Penalty [dB]	# Zeros	Penalty [dB]
1	9	6	3	7*	2	6*
2	9	3	2	4	1	4*
4	5	2	9	2	2	2
8	9	1	5	2	3	1
16	18	1	9	1	6	1

*DCIs with less than 16 DM-RS symbols, immune to our approach

8 Conclusion

Sniffing user-specific control and data traffic over 5G NR has long been considered challenging due to the concealment of crucial per-user radio interface parameters (DCI, RNTI, Scrambling ID). This paper challenges this notion. Rather than optimizing brute-force methods or relying on the clear-text RRC registration procedure as in prior works, we introduce a novel approach that exploits the linearity and non-cryptographic nature of Gold sequences to blindly recover these parameters, substantially reducing the sniffer parameter space to just a few alternatives.

Practicality is demonstrated through a concrete 5G sniffing tool capable of decoding both control and data channels, even under complex user-specific BWP configurations and real-world conditions (tested on captures from national operators). While currently limited to DL interception and not yet real-time due to prototype implementation, its algorithmic foundation, avoiding brute-force testing, suggests potential for re-engineering into operational use. To promote community collaboration and further research, we release *Golden Sniffer* open-source for academic use through our repository at https://github.com/5G-GoldenSniffer/5G-GoldenSniffer.

To counter our sniffer, we propose a simple heuristic that disrupts phase-based reference signal reconstruction by inducing artificial deep fading via cosine-squared amplitude modulation. Experiments confirm it impairs phase estimation for unauthorized receivers with minimal impact on legitimate users. Beyond immediate efficacy, this opens a promising research direction: designing future 6G air interfaces with built-in PHY-layer resilience against passive sniffing.

Ethical Considerations.

Golden Sniffer operates in a fully passive manner, without transmitting or interfering with the radio medium. All laboratory experiments were conducted in controlled testbed environments using network infrastructure and user devices under our direct control, with no involvement of external human participants. Experiments performed on commercial networks were limited to passive over-the-air observations of DL control-plane metadata and encrypted traffic only; no user-plane payloads were decrypted, decoded, or retained, and no attempt was made to identify or track users.

A Appendix: Phase Model & Assumptions

Denoting the frequency-domain received symbols by $y_{k,l}$, the model for asynchronously sampled OFDM is:

$$y_{k,l} = H_k e^{-j2\pi f_k \epsilon_l + j\varphi_l} a_{k,l} + w_{k,l} \tag{15}$$

where f_k is the k-th subcarrier frequency, ϵ_l and φ_l are, respectively, the slowly varying timing error and phase error (both assumed constant for the l-th symbol), $w_{k,l}$ is the noise and interference term, and $H_k = H(f_k)$ is the channel frequency response sampled at f_k. For a general multipath channel, the channel frequency response is given as:

$$H(f_k) = \sum_{p=1}^{L_p} \rho_p e^{-j2\pi f_k \tau_p} \tag{16}$$

where the ρ_p and τ_p represent, respectively, the baseband-equivalent amplitudes and delays of the L_p multipath components.

Assumptions. In deriving the main text expressions, two simplifying assumptions are made:

- **Dominant LOS:** In the presence of a dominant LOS term $\hat{p}$, where $|\rho_{\hat{p}}| > \sum_{p \neq \hat{p}} |\rho_p|$, the channel frequency response has no zeros and its phase is smooth and slowly varying.
- **High SNR:** In the high SNR regime, the noise and interference contribution to the phase of $y_{k,l}$ can be neglected.

Both assumptions can be easily assumed for an attacker who can decide where to position himself and use a directional antenna to listen to the gNB signal.

Under these conditions, the observed sequence of phases $\angle y_{k,l}$ can be expressed as the sum of the channel phase $\angle H_k$, the timing term $2\pi f_k \epsilon_l$, the residual carrier frequency offset φ_l, and the phase of the DM-RS modulated symbol $\angle a_{k,l}$:

$$\angle y_{k,l} \cong \angle H_k - 2\pi f_k \epsilon_l + \varphi_l + \angle a_{k,l} \tag{17}$$

Taking the difference between consecutive subcarriers' phases:

$$\angle y_{k+1,l} - \angle y_{k,l} = \angle H_{k+1} - \angle H_k + 2\pi \Delta f \epsilon_l + \angle a_{k+1,l} - \angle a_{k,l} \tag{18}$$

where Δf is the subcarrier spacing. This difference is dominated by the symbol-dependent term $\angle a_{k+1,l} - \angle a_{k,l}$, provided the above two assumptions.

Each of the three terms contributes to the observed phase increments:

- **Channel frequency response increment** $\angle H_{k+1} - \angle H_k$: assumed small and smooth due to the LOS assumption.
- **Sampling time offset** $2\pi \Delta f \epsilon_l$: small; for a fixed symbol, approximately a linear function of subcarrier index.
- **Modulation-dependent term** $\angle a_{k+1,l} - \angle a_{k,l}$: assumes only four values, multiples of $\pi/2$ due to QPSK mapping.

For typical 5G DL SNR and LOS (as in our experiments in Sec. 5), these conditions are well satisfied and justify the QPSK-based phase quantization used in the main text.

B Appendix: QPSK mapping for specific Reference Signals

PBCH DM-RS: The DM-RS sequence can be directly reconstructed, with the unknown $\bar{i}_{SSB}$. In the PBCH, one DM-RS appears every four subcarriers, resulting in 60 consecutive DM-RS samples across the second and fourth symbol.

PDCCH DM-RS: The DM-RS sequence is scrambled using the unknown 16-bit higher-layer parameter *pdcch-DMRS-ScramblingID* N_{ID}. One in every four subcarriers is dedicated to DM-RS, and the number of useful samples depends on the AL and the number of symbols. For the method to apply, there must be at least 16 consecutive samples, i.e., $AL \cdot \frac{18}{duration} \geq 16$, thus, depending on possible AL $\in \{1, 2, 4, 8, 16\}$ and PDCCH symbols $duration \in \{1, 2, 3\}$, some combinations may not provide enough samples.

PDSCH DM-RS: The DM-RS sequence can be directly reconstructed, with the unknown Scrambling ID $N_{ID}^{n_{SCID}}$, specified in the RRC Setup *DMRS-DownlinkConfig IE*. In single-layer PDSCH, one in every two subcarriers is dedicated to DM-RS, and the method requires at least 16 consecutive samples, i.e., $L_{RB} \cdot 6 \geq 16$. Therefore, the method cannot be applied for allocations smaller than 3 RBs.

NZP CSI-RS: The DM-RS sequence is scrambled using a user-specific *scramblingID* or *sequenceGenerationConfig*, both specified in the RRC Setup *NZP-CSI-RS-Resource IE*. The number of dedicated subcarriers depends on the configured density, ranging from one subcarrier every 24 (for density 0.5) to four subcarriers every 24 (for density 3). Assuming the worst condition, i.e., density 0.5, the method applies to resource grids with $N_{RB} \geq 32$.

References

[1] Olaonipekun Oluwafemi Erunkulu, Adamu Murtala Zungeru, Caspar K Lebekwe, Modisa Mosalaosi, and Joseph M Chuma. 5G Mobile Communication Applications: A Survey and Comparison of Use Cases. *IEEE Access*, 9:97251–97295, 2021.

[2] Ahmad Alalewi, Iyad Dayoub, and Soumaya Cherkaoui. On 5G-V2X Use Cases and Enabling Technologies: A Comprehensive Survey. *Ieee Access*, 9:107710–107737, 2021.

[3] Alcardo Alex Barakabitze, Arslan Ahmad, Rashid Mijumbi, and Andrew Hines. 5G Network Slicing Using SDN and NFV: A Survey of Taxonomy, Architectures and Future Challenges. *Computer Networks*, 167:106984, 2020.

[4] CJ Richards, JC McEachen, and M Tummala. RNTI Recovery Optimization Utilizing Modified Polar Decoding and Syndrome Matching. In *2024 17th International Conference on Signal Processing and Communication System (ICSPCS)*, pages 1–10. IEEE, 2024.

[5] Norbert Ludant, Pieter Robyns, and Guevara Noubir. From 5G Sniffing to Harvesting Leakages of Privacy-Preserving Messengers. In *2023 IEEE Symp. on Security and Privacy (SP)*, pages 3146–3161. IEEE, 2023.

[6] Marc Lichtman, Raghunandan Rao, Vuk Marojevic, Jeffrey Reed, and Roger Piqueras Jover. 5G NR Jamming, Spoofing, and Sniffing: Threat Assessment and Mitigation. In *2018 IEEE international conference on communications workshops (ICC Workshops)*, pages 1–6. IEEE, 2018.

[7] 3GPP. Evolved Universal Terrestrial Radio Access (E-UTRA); Physical Channels and Modulation. Technical Specification (TS) 36.211, 3rd Generation Partnership Project (3GPP), 04 2024. 18.0.1.

[8] Shijie Luo, Garbelini Matheus E, Chattopadhyay Sudipta, and Jianying Zhou. Sni5Gect: A Practical Approach to Inject aNRchy into 5G NR. In *34th USENIX Security Symposium (USENIX Security 25)*, 2025.

[9] Haoran Wan, Xuyang Cao, Alexander Marder, and Kyle Jamieson. NR-Scope: A Practical 5G Standalone Telemetry Tool. In *Proceedings of the 20th International Conference on emerging Networking EXperiments and Technologies*, pages 73–80, 2024.

[10] Swarun Kumar, Ezzeldin Hamed, Dina Katabi, and Li Erran Li. LTE Radio Analytics Made Easy and Accessible. *ACM SIGCOMM Computer Communication Review*, 44(4):211–222, 2014.

[11] Xiufeng Xie, Xinyu Zhang, Swarun Kumar, and Li Erran Li. piStream: Physical Layer Informed Adaptive Video Streaming over LTE. In *Proceedings of the 21st Annual International Conference on Mobile Computing and Networking*, pages 413–425, 2015.

[12] Robert Falkenberg, Christoph Ide, and Christian Wietfeld. Client-Based Control Channel Analysis for Connectivity Estimation in LTE Networks. In *2016 IEEE 84th Vehicular Technology Conference (VTC-Fall)*, pages 1–6. IEEE, 2016.

[13] Nicola Bui and Joerg Widmer. OWL: A Reliable Online Watcher for LTE Control Channel Measurements. In *Proceedings of the 5th Workshop on All Things Cellular: Operations, Applications and Challenges*, pages 25–30, 2016.

[14] Robert Falkenberg and Christian Wietfeld. FALCON: An Accurate Real-Time Monitor for Client-Based Mobile Network Data Analytics. In *2019 IEEE Global Communications Conference (GLOBECOM)*, pages 1–7. IEEE, 2019.

[15] Martin Kotuliak, Simon Erni, Patrick Leu, Marc Röschlin, and Srdjan Čapkun. LTrack: Stealthy Tracking of Mobile Phones in LTE. In *31st USENIX Security Symposium (USENIX Security 22)*, pages 1291–1306, 2022.

[16] Tuan Dinh Hoang, CheolJun Park, Mincheol Son, Taekkyung Oh, Sangwook Bae, Junho Ahn, Beomseok Oh, and Yongdae Kim. LTESniffer: An Open-Source LTE Downlink/Uplink Eavesdropper. In *Proceedings of the 16th ACM Conference on Security and Privacy in Wireless and Mobile Networks*, pages 43–48, 2023.

[17] Vuk Marojevic, Raghunandan M Rao, Sean Ha, and Jeffrey H Reed. Performance Analysis of a Mission-Critical Portable LTE System in Targeted RF Interference. In *2017 IEEE 86th Vehicular Technology Conference (VTC-Fall)*, pages 1–6. IEEE, 2017.

[18] Mina Labib, Vuk Marojevic, Jeffrey H Reed, and Amir I Zaghloul. Enhancing the Robustness of LTE Systems: Analysis and Evolution of the Cell Selection Process. *IEEE Communications Magazine*, 55(2):208–215, 2017.

[19] Marc Lichtman, Roger Piqueras Jover, Mina Labib, Raghunandan Rao, Vuk Marojevic, and Jeffrey H Reed. LTE/LTE-A Jamming, Spoofing, and Sniffing: Threat Assessment and Mitigation. *IEEE Communications Magazine*, 54(4):54–61, 2016.

[20] Marc Lichtman, Jeffrey H Reed, T Charles Clancy, and Mark Norton. Vulnerability of LTE to Hostile Interference. In *2013 IEEE Global Conference on Signal and Information Processing*, pages 285–288. Ieee, 2013.

[21] Andrea Paci, Gabriele Bologna, Ivan Palamà, and Giuseppe Bianchi. FlashCatch: Minimizing Disruption in IMSI Catcher Operations. In *18th ACM Conference on Security and Privacy in Wireless and Mobile Networks*, WiSec 2025, page 124–135, New York, NY, USA, 2025. Association for Computing Machinery.

[22] Shinjo Park. *Why We Cannot Win: On Fake Base Stations and Their Detection Methods*. PhD thesis, Technische Universität Berlin, 2023.

[23] Ivan Palamà, Francesco Gringoli, Giuseppe Bianchi, and Nicola Blefari-Melazzi. IMSI Catchers in the Wild: A Real World 4G/5G Assessment. *Computer Networks*, 194:108137, 2021.

[24] Syed Rafiul Hussain, Mitziu Echeverria, Omar Chowdhury, Ninghui Li, and Elisa Bertino. Privacy Attacks to the 4G and 5G Cellular Paging Protocols Using Side Channel Information. *Network and distributed systems security (NDSS) symposium 2019*, 2019.

[25] Altaf Shaik, Ravishankar Borgaonkar, Shinjo Park, and Jean-Pierre Seifert. New Vulnerabilities in 4G and 5G Cellular Access Network Protocols: Exposing Device Capabilities. In *Proceedings of the 12th Conference on Security and Privacy in Wireless and Mobile Networks*, pages 221–231, 2019.

[26] Simon Erni, Martin Kotuliak, Patrick Leu, Marc Roeschlin, and Srdjan Capkun. AdaptOver: Adaptive Overshadowing Attacks in Cellular Networks. In *Proceedings of the 28th Annual International Conference on Mobile Computing And Networking*, pages 743–755, 2022.

[27] Hojoon Yang, Sangwook Bae, Mincheol Son, Hongil Kim, Song Min Kim, and Yongdae Kim. Hiding in Plain Signal: Physical Signal Overshadowing Attack on {LTE}. In *28th USENIX Security Symposium (USENIX Security 19)*, pages 55–72, 2019.

[28] David Rupprecht, Katharina Kohls, Thorsten Holz, and Christina Pöpper. Breaking LTE on Layer Two. In *2019 IEEE Symposium on Security and Privacy (SP)*, pages 1121–1136. IEEE, 2019.

[29] Michal Harvanek, Jan Bolcek, Jan Kufa, Ladislav Polak, Marek Simka, and Roman Marsalek. Survey on 5G Physical Layer Security Threats and Countermeasures. *Sensors (Basel, Switzerland)*, 24(17):5523, 2024.

[30] Norbert Ludant, Marinos Vomvas, and Guevara Noubir. Unprotected 4G/5G Control Procedures at Low Layers Considered Dangerous. *arXiv preprint arXiv:2403.06717*, 2024.

[31] Mohamad Saalim Wani, Michael Rademacher, Thorsten Horstmann, and Mathias Kretschmer. Security Vulnerabilities in 5G Non-Stand-Alone Networks: A Systematic Analysis and Attack Taxonomy. *Journal of Cybersecurity and Privacy*, 4(1):23–40, 2024.

[32] Norbert Ludant and Guevara Noubir. SigUnder: A Stealthy 5G Low Power Attack and Defenses. In *Proceedings of the 14th ACM Conference on Security and Privacy in Wireless and Mobile Networks*, pages 250–260, 2021.

[33] Youness Arjoune and Saleh Faruque. Smart Jamming Attacks in 5G New Radio: A Review. In *2020 10th Annual Computing and Communication Workshop and Conference (CCWC)*, pages 1010–1015. IEEE, 2020.
[34] Keysight. WaveJudge Wireless Analyzer Solutions, 2025. Accessed on 2025-03-18.
[35] Qualcomm. QXDM Professional™ Tool, 2020. Accessed on 2025-03-18.
[36] Actix. Actix Analyzer, 2020. Accessed on 2025-03-18.
[37] Infovista. TEMS™ Investigation | Mobile Network Drive Testing, 2025. Accessed on 2025-03-18.
[38] thinkRF. Autonomous & Continuous Spectrum Intelligence Platform, 2025. Accessed on 2025-03-18.
[39] Sangwoo Lee, Liuyi Jin, and Radu Stoleru. Saflo: eBPF-Based MPTCP Scheduler for Mitigating Traffic Analysis Attacks in Cellular Networks, 2025.
[40] Gunwoo Yoon and Byeongdo Hong. Scalable and Robust Mobile Activity Fingerprinting via Over-the-Air Control Channel in 5G Networks, 2024. Submitted on 19 September 2024.
[41] SRS. Open source O-RAN 5G CU/DU solution from Software Radio Systems (SRS), 2025. Accessed on 2025-03-18.
[42] 3GPP. NR; Physical Channels and Modulation. Technical Specification (TS) 38.211, 3rd Generation Partnership Project (3GPP), 07 2024. 18.3.0.
[43] 3GPP. NR; Base Station (BS) Radio Transmission and Reception. Technical report (TR) 38.104, 3rd Generation Partnership Project (3GPP), 06 2025. 19.1.0.
[44] 3GPP. NR; Multiplexing and Channel Coding. Technical Specification (TS) 38.212, 3rd Generation Partnership Project (3GPP), 07 2023. 18.3.0.
[45] Sukchan Lee. Open5GS, 2025. Accessed on 2025-03-18.
[46] Rohde & Schwarz. R&S®FSVA3000 Signal and spectrum analyzer, 2025.
[47] Ettus Research. *USRP™ N310 Simplifying SDR Deployment.*
[48] 3GPP. Study on Channel Model for Frequencies from 0.5 to 100 GHz. Technical report (TR) 38.901, 3rd Generation Partnership Project (3GPP), 07 2025. 19.0.0.